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Course for the Faculties of Sciences and Chemical Sciences

Álvaro Lozano Murciego and André Sales Mendes

Universidad de Salamanca

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CONTENTS

I	Quadratic Equations	3
1	Quadratic Equations	5
1.1	The discriminant	5
1.2	General formula	5
1.3	Worked example	6
1.4	Interpretation	6
II	Higher Degree Equations	7
2	Higher Degree Equations	9
2.1	What are higher degree equations?	9
2.2	Solving degree 3 and 4 equations	9
2.3	The jump: degree 5 and beyond	10
2.4	How do we find solutions then?	11
2.5	Summary	11
2.6	Final reflection	12
III	Research paper	13
3	Summary of the paper “Transverse expansion of the metric at null infinity”	15
3.1	Objective	15
3.2	Main results	15
3.3	Method	15
3.4	Obstacles and contributions	16
3.5	Relevance	16
4	Bibliography	17

This version of the book is focused on a single new chapter about quadratic equations.

What you will find here

- Definition of the quadratic equation
- The general formula and the discriminant
- How to solve a quadratic equation step by step
- A worked example with interpretation
- *Go to the quadratic equations chapter*

Note

The rest of the book content has been hidden to show only this new chapter.

Part I

Quadratic Equations

QUADRATIC EQUATIONS

A quadratic equation is an equality of the form:

$$ax^2 + bx + c = 0$$

where:

- a , b and c are real numbers.
- $a \neq 0$.

1.1 The discriminant

The discriminant is:

$$\Delta = b^2 - 4ac$$

It defines three cases:

- $\Delta > 0 \rightarrow$ two distinct real solutions.
- $\Delta = 0 \rightarrow$ one repeated real solution.
- $\Delta < 0 \rightarrow$ no real solutions.

1.2 General formula

The solutions are given by:

$$x = \frac{-b \pm \sqrt{\Delta}}{2a}$$

1.2.1 How to solve it

1. Identify the coefficients a , b and c .
2. Compute the discriminant $\Delta = b^2 - 4ac$.
3. Check the sign of Δ .
4. Substitute into the general formula.

1.3 Worked example

Solve:

$$2x^2 - 3x - 2 = 0$$

Here $a = 2$, $b = -3$ and $c = -2$.

Compute the discriminant:

$$\Delta = (-3)^2 - 4 \cdot 2 \cdot (-2) = 9 + 16 = 25$$

Substitute into the formula:

$$x = \frac{-(-3) \pm \sqrt{25}}{2 \cdot 2} = \frac{3 \pm 5}{4}$$

The two solutions are:

- $x_1 = \frac{3+5}{4} = 2$
- $x_2 = \frac{3-5}{4} = -\frac{1}{2}$

1.4 Interpretation

- A quadratic equation defines a parabola.
- If $\Delta > 0$, the parabola crosses the x -axis in two points.
- If $\Delta = 0$, the parabola touches the x -axis at one point.
- If $\Delta < 0$, the parabola does not cross the x -axis.

Tip

Verify the solutions by substituting them back into the original equation.

Part II

Higher Degree Equations

HIGHER DEGREE EQUATIONS

2.1 What are higher degree equations?

In previous chapters, you studied quadratic equations, which have the general form:

$$ax^2 + bx + c = 0$$

Higher degree equations are polynomials of degree 3 or greater. Remember that the **degree** of a polynomial is the highest exponent of the variable.

The most common general forms are:

- **Degree 3 (cubic equations):** $ax^3 + bx^2 + cx + d = 0$
- **Degree 4 (quartic equations):** $ax^4 + bx^3 + cx^2 + dx + e = 0$
- **Degree 5 (quintic equations):** $ax^5 + bx^4 + cx^3 + dx^2 + ex + f = 0$

2.2 Solving degree 3 and 4 equations

2.2.1 Cubic equations

For degree 3 equations, **there is a closed formula** that allows you to calculate the exact roots. It is much more complicated than the formula for quadratic equations, but it exists.

Cardano's formula (16th century) allows you to find the exact solutions of any cubic equation. However, it is so complicated that in practice it is rare to use it by hand:

Historical curiosity

Cardano's formula was a revolutionary discovery during the Italian Renaissance. Although mathematicians like Tartaglia and Cardano developed these methods, the formula remains difficult to apply without computational help.

2.2.2 Quartic equations

For degree 4 equations, there is also a **closed formula** (Ferrari's formula, also from the 16th century). As with cubics, it is very complicated, but it exists.

Practical methods

Instead of using closed formulas for degrees 3 and 4, modern mathematicians and engineers usually:

- Factor if possible
- Use iterative numerical methods (Newton-Raphson, bisection, etc.)
- Use computational tools like Python, MATLAB, or Mathematica

2.3 The jump: degree 5 and beyond

2.3.1 The Abel-Ruffini theorem

Here comes a **surprising and fundamental result**: for degree 5 or higher equations, **there is no closed formula** that uses only arithmetic operations and roots.

This result, proved in the early 19th century by **Paolo Ruffini** and **Niels Henrik Abel**, is known as the **Abel-Ruffini Theorem** or **Impossibility Theorem**.

Abel-Ruffini Theorem

For polynomial equations of degree 5 or higher, **there is no closed formula** that expresses the solutions only by means of addition, subtraction, multiplication, division, and taking n th roots of the coefficients.

2.3.2 What does this mean?

Although we know that every polynomial equation of degree n has exactly n roots (including complex ones), **we cannot write a general formula** that gives us all of them.

This does not mean that the roots do not exist or that we cannot find them. It only means that we cannot express them using a “formula” as we do with second degree equations.

2.3.3 Conceptual example

For a cubic equation like $x^3 - 2 = 0$, the solution is $x = \sqrt[3]{2}$ (the real cube root of 2). We can express this solution exactly.

But for a degree 5 equation like $x^5 + x + 1 = 0$, **there is no way to write the exact solution** in terms of addition, subtraction, multiplication, division, and roots.

2.4 How do we find solutions then?

Although there is no closed formula, we have **alternative methods**:

2.4.1 1. Numerical methods

We use iterative algorithms that approximate solutions to any desired precision:

- **Newton-Raphson method**: starting from an initial approximation, it iteratively improves the solution
- **Bisection method**: divides the interval containing the root in half, repeating until it finds it
- **Modern methods**: sophisticated algorithms implemented in Python, MATLAB, etc.

```
import numpy as np
from scipy.optimize import fsolve

# Find roots of  $x^5 + x + 1 = 0$ 
def f(x):
    return x**5 + x + 1

# Initial approximation
x0 = -1.0

# Solve
solution = fsolve(f, x0)
print(f"Approximate solution: {solution[0]:.6f}")
```

2.4.2 2. Special algebraic methods

For certain degree 5 or higher equations with special structure, sometimes ingenious algebraic techniques can be used to find the roots.

2.4.3 3. Graphical methods

Drawing the function and seeing where it crosses the x -axis is a visual way to approximate solutions.

2.5 Summary

Degree	Is there a closed formula?	How to solve?
1	✓ Yes	Simple algebra
2	✓ Yes (Bhaskara)	Quadratic formula
3	✓ Yes (Cardano)	Cardano's formula or numerical methods
4	✓ Yes (Ferrari)	Ferrari's formula or numerical methods
5+	✗ No (Abel-Ruffini)	Numerical methods or special algebraic techniques

2.6 Final reflection

The discovery that there is no closed formula for degree 5 and higher was a **humbling lesson** for mathematics. It showed that not all natural questions have answers in the form we expect.

However, this “does not exist” led to the development of:

- **Advanced numerical methods** that work for any polynomial
- **Galois theory**, a profound branch of mathematics that studies why certain polynomials do not have solutions with radicals
- **Computational tools** that can solve practically any equation to the precision we need

In modern practice, we rarely need to memorize formulas. We use **software to solve equations of any degree**, but it is important to understand why that is necessary.

Part III

Research paper

SUMMARY OF THE PAPER “TRANSVERSE EXPANSION OF THE METRIC AT NULL INFINITY”

This chapter summarizes the article available in `resources/Paper_scri.pdf`, written by Marc Mars and Gabriel Sánchez-Pérez.

3.1 Objective

The paper analyzes the conformal Einstein equations at null infinity (I) to all orders without imposing restrictions on spacetime dimension, the topology of I , or the fall-off conditions for the Weyl tensor.

3.2 Main results

- It identifies the free data at null infinity that determine the asymptotic geometry.
- It proves that two asymptotically flat spacetimes with the same free data at I are isometric to infinite order.
- It establishes an existence theorem for asymptotically flat spacetimes realizing prescribed initial data at I .
- It introduces a detached definition of null infinity that does not rely on a preexisting spacetime.

3.3 Method

- A coordinate-free approach is used.
- The conformal factor Ω is treated as a dynamical variable.
- The paper uses null hypersurface data and the transverse expansion of the metric.

3.4 Obstacles and contributions

- Two geometric obstructions appear: the Coulombian obstruction tensor and the radiative obstruction tensor.
- The free data structure depends on spacetime dimension and parity.
- This work extends classical Bondi-style analyses and Fefferman–Graham expansions to non-spherical topologies and general dimensions.

3.5 Relevance

The chapter highlights that the geometry of null infinity can be described in a much more general conformal framework, which is useful for studying gravitational radiation and asymptotically flat fields in diverse settings.

3.5.1 Note

The original text is in `resources/Paper_scri.pdf` and this page summarizes its main ideas.

BIBLIOGRAPHY

This page collects every reference cited in the book.

Additional recommended references on quadratic and higher-degree equations, as well as the Mars and Sánchez-Pérez scientific article.